UNIVERSITY OF THE PEOPLE

BIOL 1301-01 - AY2026-T5

Assignment 5

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Mendelian Genetics, DNA Replication, Gene Expression, and CRISPR-Cas9 in Drosophila melanogaster

Introduction

Genetics explains how traits pass from parents to offspring. Mendel's laws of inheritance, the molecular structure of DNA, and the processes of replication, transcription, and translation together explain how genotypes become phenotypes. Modern tools such as CRISPR-Cas9 now allow scientists to test and refine these classical predictions directly in living organisms, including Drosophila melanogaster.

1. Dihybrid Punnett Square and Phenotypic Ratio

Parent 1 (RrNn) is heterozygous for both eye color and wing shape and therefore produces four gamete types: RN, Rn, rN, and rn. Parent 2 (Rrnn) is heterozygous for eye color but homozygous recessive for wing shape, so it produces only two gamete types: Rn and rn. Combining these gametes in the dihybrid Punnett square below yields eight offspring combinations across six unique genotypes.

P2 ↓ / P1 →	**RN**	**Rn**	**rN**	**rn**
Rn	RRNn	RRnn	RrNn	Rrnn
rn	RrNn	Rrnn	rrNn	rrnn

Grouping these eight offspring by observable phenotype produces the following distribution:

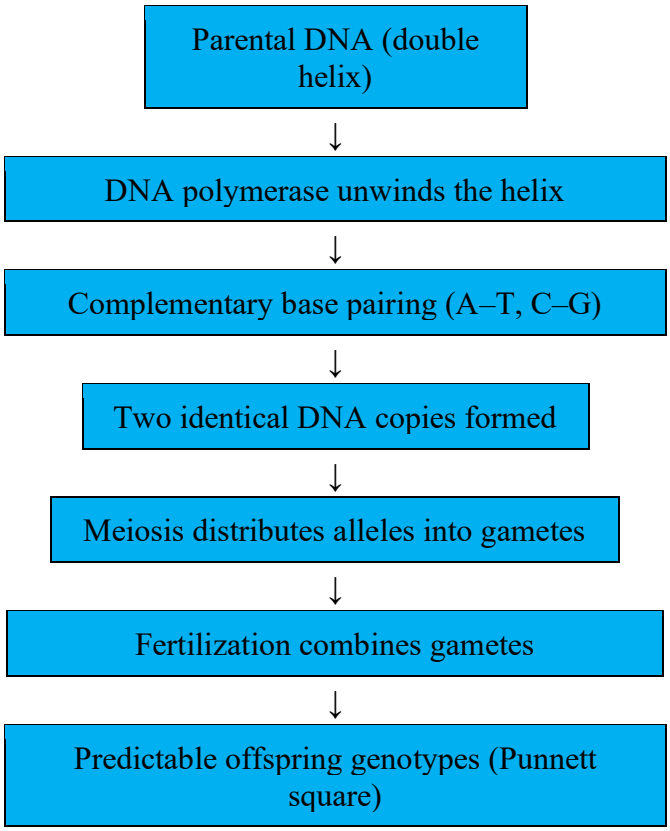
Phenotype	**Count (out of 8)**
Red eyes, Normal wings (R_N_)	3
Red eyes, Vestigial wings (R_nn)	3
White eyes, Normal wings (rrN_)	1
White eyes, Vestigial wings (rrnn)	1

The resulting phenotypic ratio is 3 : 3 : 1 : 1 (red/normal : red/vestigial : white/normal : white/vestigial). This outcome reflects Mendel's law of segregation, since each parent transmits only one allele per gene to every gamete produced (Clark et al., 2023). Considered separately, the eye-color trait follows the classic 3:1 ratio expected from an Rr × Rr cross, while the wing-shape trait follows a 1:1 ratio expected from an Nn × nn cross. Multiplying these two independent ratios together, in keeping with the law of independent assortment, produces the combined 3:3:1:1 result shown above.

2. Base-Pairing in DNA Replication

The predictable outcomes shown in the Punnett square depend on DNA being copied with near-perfect accuracy before meiosis. During replication, the enzyme DNA polymerase unwinds the double helix and synthesizes a new complementary strand for each original strand, guided by strict base-pairing rules in which adenine bonds only with thymine and cytosine bonds only with guanine (Clark et al., 2023). Because each nucleotide can pair correctly with only one complementary base, replication errors are rare, and the alleles present in a parent's cells are passed faithfully into the gametes that parent produces. This fidelity is what allows the phenotypic ratios calculated from a Punnett square to be treated as reliable biological predictions rather than coincidence.

Flowchart: DNA Replication and Predictable Inheritance

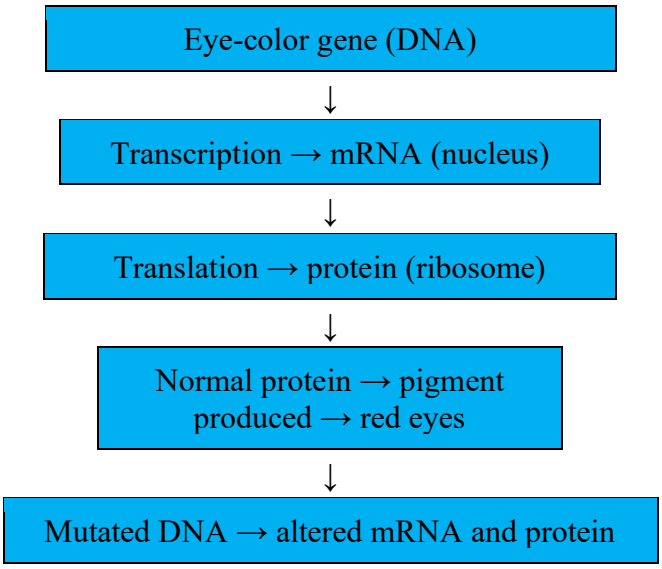


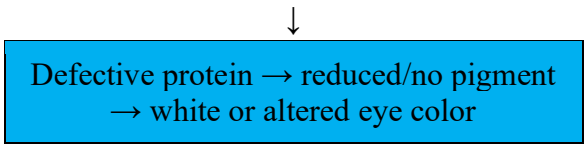
3. Transcription, Translation, and the Effect of Mutation

Phenotypes are the visible result of proteins encoded by genes. Transcription copies a gene's DNA sequence into messenger RNA (mRNA) within the nucleus using RNA polymerase. Translation then occurs at the ribosome, where transfer RNA (tRNA) delivers amino acids that are joined into a polypeptide chain in the order specified by the mRNA sequence (OpenStax, 2023). In Drosophila, the protein produced from the red-eye-color gene deposits pigment into the eye during development, producing the red-eyed phenotype predicted by the Punnett square.

A mutation in this gene's DNA sequence can alter the mRNA codon sequence and, in turn, change the amino acid sequence of the resulting protein. If the altered protein can no longer synthesize or transport pigment correctly, the fly's eyes may appear white, orange, or another unexpected shade, even though inheritance alone predicted a red-eyed phenotype (Griffiths et al., 2020). This shows that inheritance patterns describe which alleles are transmitted, while gene expression and mutation ultimately determine whether those alleles function as expected.

Flowchart: Transcription, Translation, and Mutation Effects





4. CRISPR-Cas9 in Drosophila Research

CRISPR-Cas9 allows researchers to cut DNA at a specific location selected by a guide RNA, then disrupt, delete, or replace the targeted sequence (Alberts et al., 2022). In Drosophila research, scientists commonly use CRISPR-Cas9 to knock out or edit the eye-color gene directly and then compare the resulting flies to unedited controls, confirming the gene's precise function rather than inferring it solely from crosses.

Because CRISPR can target a single gene while leaving the rest of the genome unchanged, it also helps explain why real crosses sometimes deviate from the ratios a simple Punnett square predicts. Gene linkage, incomplete dominance, epistasis between interacting genes, and spontaneous mutations can all cause observed phenotypes to differ from expectations. By selectively editing one candidate gene at a time, researchers can isolate which specific factor is responsible for an unexpected inheritance pattern, turning CRISPR into a tool that tests, rather than merely assumes, Mendelian predictions.

Conclusion

Mendelian inheritance, accurate DNA replication, and gene expression together explain how Drosophila offspring acquire predictable eye color and wing shape. Mutations in the DNA sequence can, however, disrupt this chain by altering the proteins that genes encode. CRISPR-Cas9 gives researchers a precise way to test these relationships directly, confirming gene function and clarifying why inherited traits do not always match textbook expectations.

References

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